

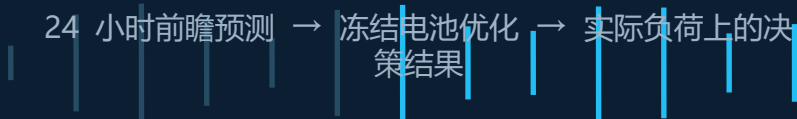
DOEF

让负荷预测服务于 储能削峰决策

面向储能决策的集成负荷预测方法



预测是输入，不是终点

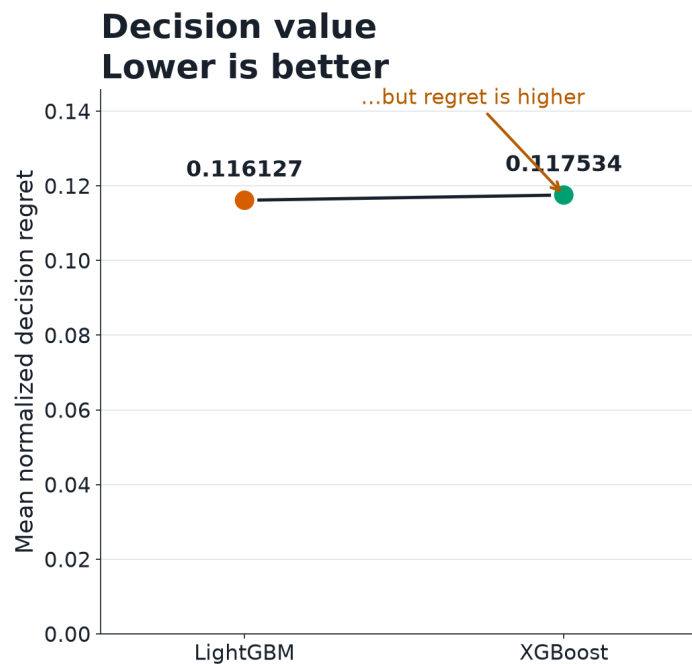
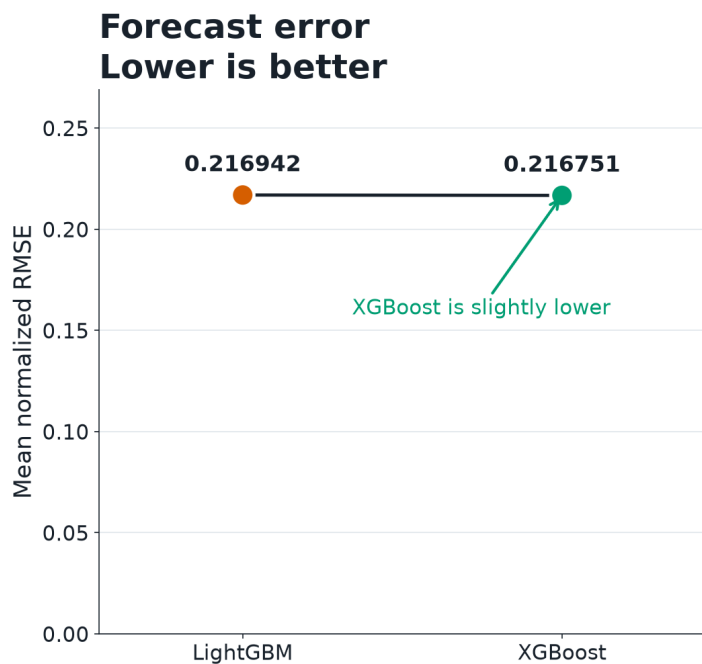


更低的预测 RMSE，并未带来更低的储能决策 regret

冻结八栋建筑 Test 汇总中，XGBoost 与 LightGBM 在预测误差与决策 regret 上出现排序反转。

Forecast metrics alone do not determine decision value

Frozen eight-building Test aggregate · equal-building mean after Train-mean normalization



Directional counterexample only: the gap is small and project-scoped; it does not imply that forecast accuracy is unimportant.

核心矛盾

RMSE

XGBoost
0.216751

Decision regret

LightGBM
0.116127

同一个“更准”
不一定意味着同一个“更好调度”

范围内反例 ≠ 预测精度无用

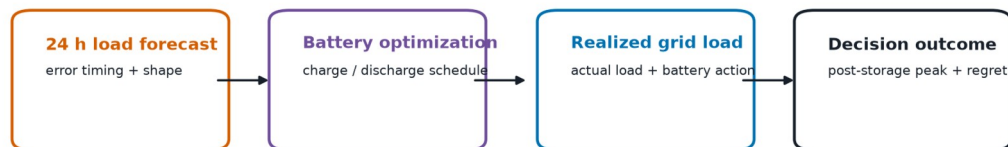
Frozen source: outputs/phase13_1/figures/sci_01_forecast_decision_mismatch.png · Phase 9 final benchmark

预测只是中间环节，调度结果才是工程终点

负荷预测会改变储能充放电计划；最终评价发生在真实负荷与电池动作叠加后的结果上。

Prediction is an input to the decision — not the endpoint

The same scalar forecast score can hide different errors at decision-critical hours



Evaluate both forecast quality and downstream storage value

CONCEPTUAL / EXPLANATORY · no experimental values encoded · no cost, carbon, or energy-savings claim

CONCEPTUAL / EXPLANATORY

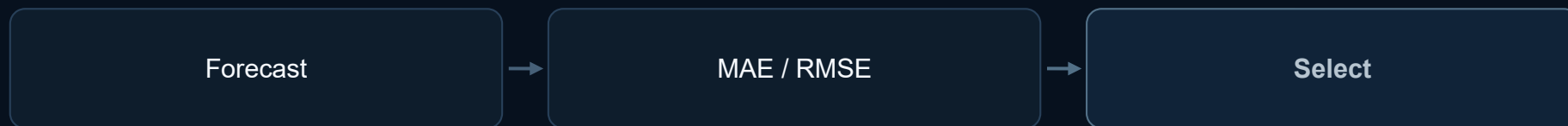
所有预测方法进入同一套冻结电池约束，保证比较的是“预测带来的决策差异”，不是不同优化器。

Conceptual asset: outputs/phase13_1/presentation_assets/con_01_prediction_decision_bridge.png

模型不只要预测得准，还要在 Validation 上支持更好的储能决策

保持预测组件与电池约束不变，只改变“选择依据”：从预测误差，转向 Validation 下游决策 regret。

传统预测选择



预测误差最小

DOEF 选择



决策 regret 最小

唯一强调的改变：SELECTION OBJECTIVE

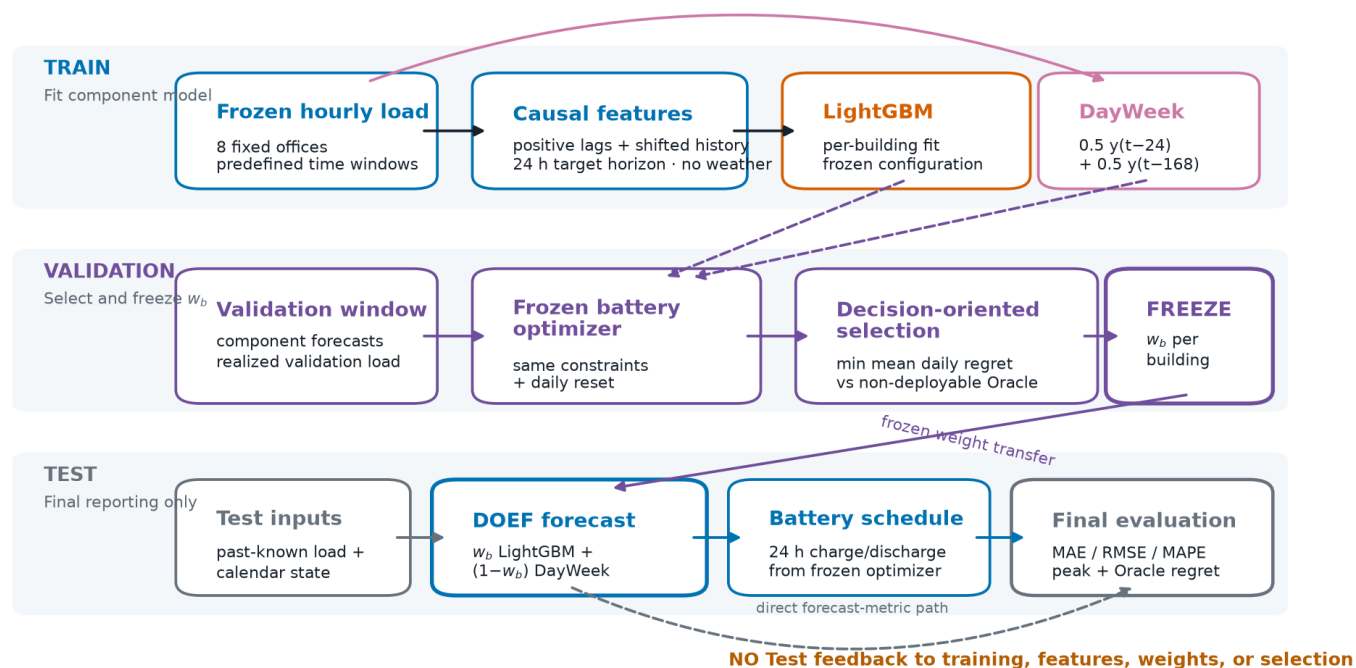
不是联合训练 · 不是新基础模型 · 不是用 Test 调权重

DOEF 在 Validation 上按决策 regret 冻结权重，再进入 Test

每栋建筑的 DayWeek-LightGBM 权重只在 Validation 选择；Test 对训练、权重、tie-break 和电池配置没有反馈路径。

DOEF v1.0 — frozen causal architecture

Prediction serves the storage decision · Validation selects · Test evaluates only



ALGORITHM FROZEN · deterministic ensemble and constrained optimization · not joint training, bilevel optimization, online learning, or a deployable Oracle.

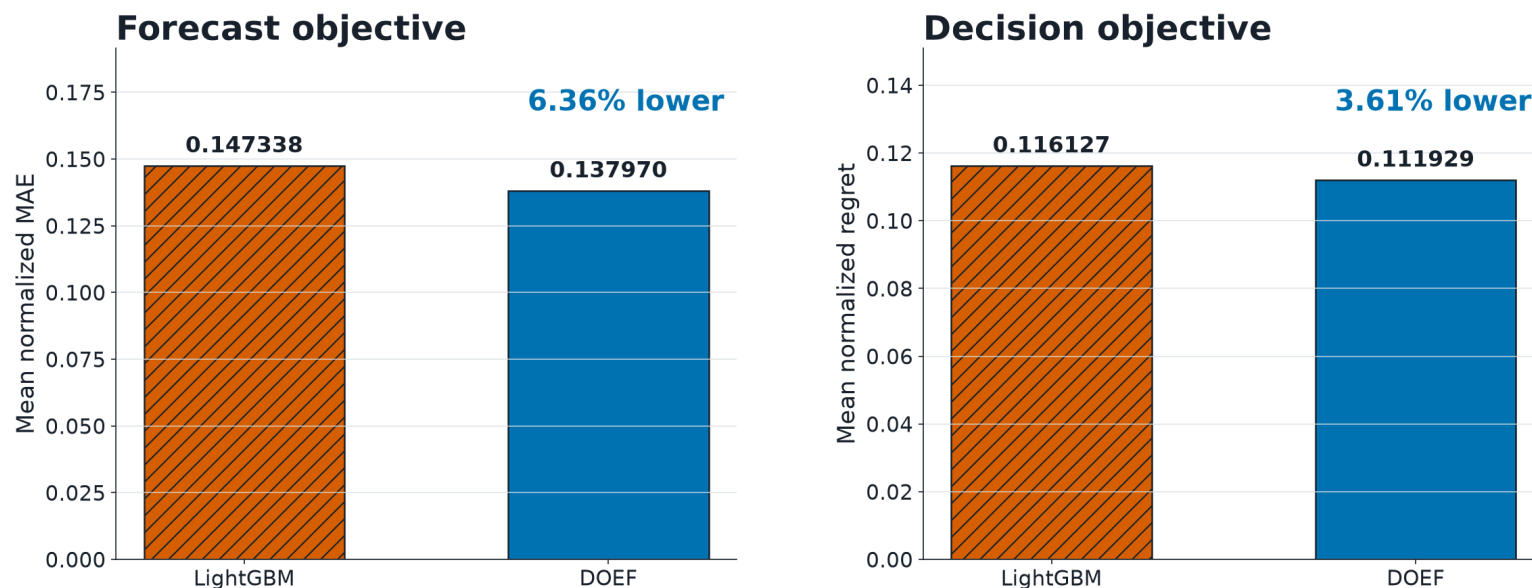
Frozen method visual: outputs/phase13_1/architecture/doef_architecture_master.png · formula: $w_b \cdot \text{LightGBM} + (1-w_b) \cdot \text{DayWeek}$

DOEF 同时降低预测误差与储能决策 regret

对比冻结 LightGBM：八栋建筑等权聚合的 normalized MAE 与 normalized decision regret 均下降。

DOEF improves both frozen aggregate objectives

Frozen eight-building Test aggregate · comparison against LightGBM · lower is better



Building-level decision regret: 6 wins · 2 ties · 0 losses

Peak-shaving context: mean 6.44%; minimum -1.02% (negative case retained). Oracle-relative regret is not monetary regret.

6.36%

normalized MAE 相对改善

3.61%

normalized regret 相对改善

6 / 2 / 0

win / tie / loss

负峰值案例保留: min = -1.017%

Frozen source: outputs/phase13_1/figures/sci_02_doef_main_results.png · Oracle-relative regret ≠ monetary regret

RMSE 衡量预测误差； regret 衡量误差经过调度后的工程后果

两个指标回答不同问题，因此在项目范围内不保证给出相同的模型排序。

FORECAST METRIC

RMSE

“预测偏了多少？”

对所有时间点的误差做平方惩罚并聚合。
它是必要的预测质量检查，但不是储能决策本身。

不同终点
→

DOWNSTREAM METRIC

regret

“这次预测让调度损失了多少？”

预测 → 电池优化 → 实际负荷，
再与不可部署 Oracle 的最优决策结果比较。

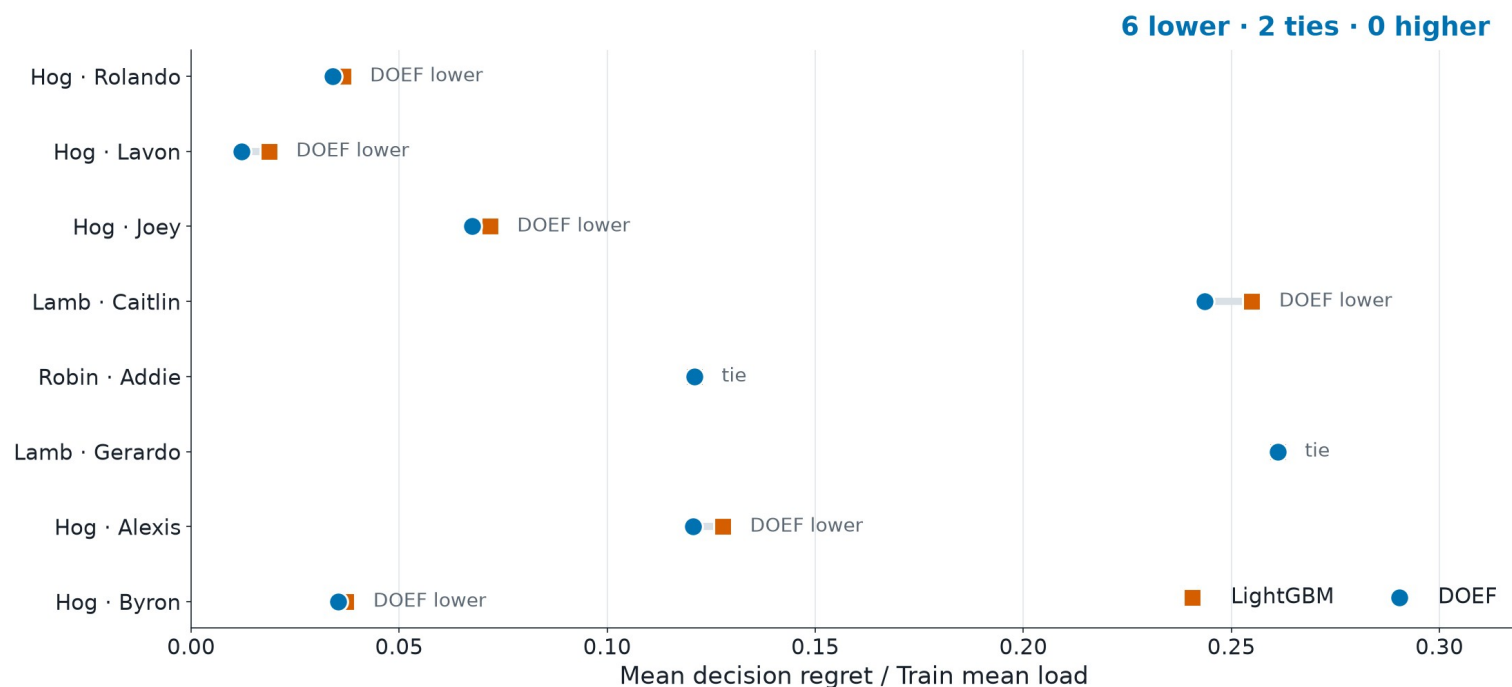
结论：两者都要看，不把一个替代另一个

全部八栋固定建筑均被展示：六胜、两平、零负

建筑在结果评估前冻结：296 个 office 候选 → 82 个质量通过 → 仅用 Train 期形态预选 8 栋。

Decision value across all eight fixed buildings

Frozen Test mean decision regret normalized by each building's frozen Train mean load · lower is better



All eight preselected offices are shown in frozen selection order. This is multi-building robustness, not unseen-building transfer.

Frozen source: SCI-03 · buildings preserved in preselected frozen order; no sorting/deletion

8 / 8

全部展示

6

lower

2

ties

0

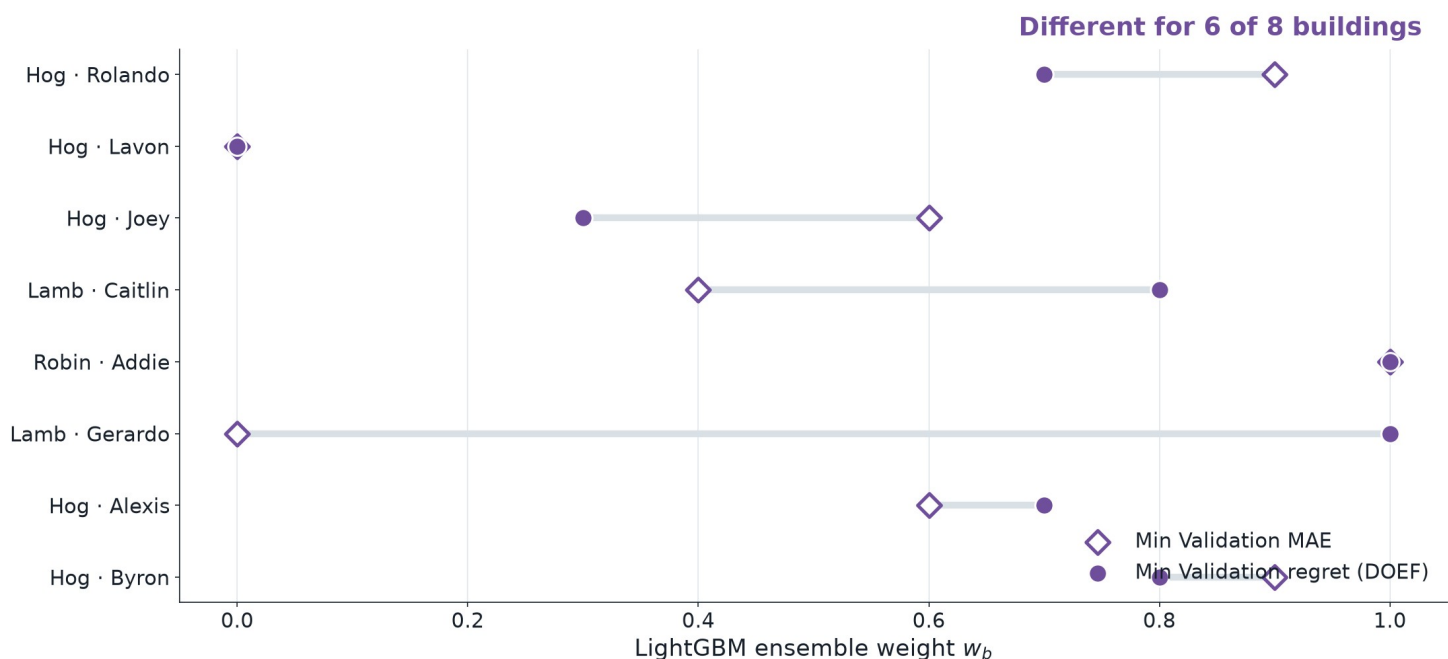
higher

权重在 Validation 冻结, Test 不参与选择

11 个候选权重在预先声明的 Validation 网格上选择; Test 只用于最终评价, 不反馈到模型、权重或 tie-break。

Validation objectives select different ensemble weights

Per-building LightGBM weight; remaining weight is DayWeek · selection uses Validation only



Test is excluded from weight and tie-break selection. Points show frozen Phase 8 weights; no optimization was rerun.

NO TEST-TIME TUNING

Validation

搜索 0.0 → 1.0
步长 0.1



FREEZE w_p

Test

evaluation only

Frozen source: SCI-04 · candidate grid = 11 weights · prior historical Test visibility means "pristine blind test" is not claimed

冻结流程明确了次日预测如何生成储能调度，但尚非现场部署

这是可复现的 next-day operating sequence，而不是已在真实楼宇上线的控制系统。



REPEATABLE

FROZEN

NOT DEPLOYED

工程价值：把 AI 预测、储能约束、决策评价连成一条可审计链。

创新不在基础预测器，而在把模型选择对齐储能决策

DOEF 的贡献是可审计的 decision-oriented framework，而不是把 LightGBM 或简单加权包装成新模型。

Four verified contribution pillars

01 Causal forecasting

24 h horizon
positive lags + shifted history
no future-target leakage

02 Decision-oriented ensemble

Validation regret selects w_b
LightGBM + DayWeek
Test excluded from selection

03 Multi-building robustness

8 fixed heterogeneous offices
Train-only morphology sampling
not unseen-building transfer

04 Storage decision evaluation

same constrained optimizer
realized peaks + Oracle regret
alongside forecast metrics

CONCEPTUAL / EXPLANATORY · contribution is project-scoped; no first / SOTA / end-to-end claim

Conceptual asset: CON-02 · no first/world-first/SOTA/end-to-end deployment claim

让预测评价 与最终储能决策对齐

FOR BATTERY PEAK SHAVING

预测模型不仅要“预测得准”，也要验证它是否支持更好的工程决策。

DOEF v1.0 • frozen evidence

八栋固定办公建筑 · 单一未来 Test 月 · 24 h horizon · 无成本 / 碳 / 现场部署结论

01 预测误差 $\neq$ 决策价值

RMSE 与 regret 回答不同问题

02 Validation 对齐选择

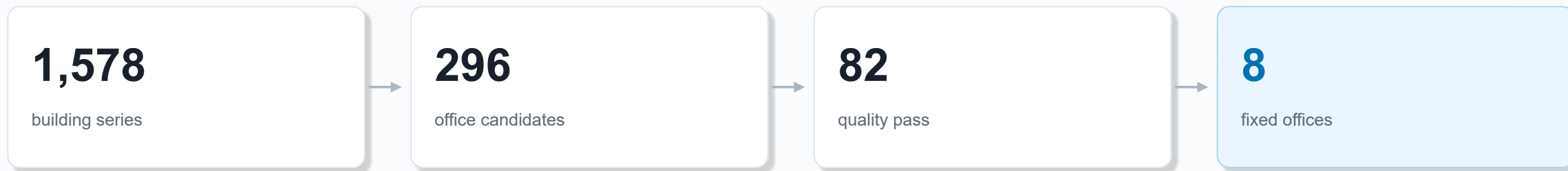
按储能 regret 冻结 w_β

03 Test 只做最终评价

固定 8 栋 · 全部结果保留

建筑范围在结果评估前冻结

质量筛选 + Train-only 负荷形态覆盖, Validation/Test 模型或决策性能不参与建筑预选。



冻结建筑顺序

01 Hog · Rolando

05 Robin · Addie

02 Hog · Lavon

06 Lamb · Gerardo

03 Hog · Joey

07 Hog · Alexis

04 Lamb · Caitlin

08 Hog · Byron

Train-only morphology features

- log mean load
- peak-to-average ratio
- lag-24 autocorrelation
- weekday-weekend difference
- robust outlier fraction
- coefficient of variation
- peak-valley range
- lag-168 autocorrelation
- daily-profile variability

NO Validation/Test performance used

Frozen source: outputs/phase7/selected_buildings.json

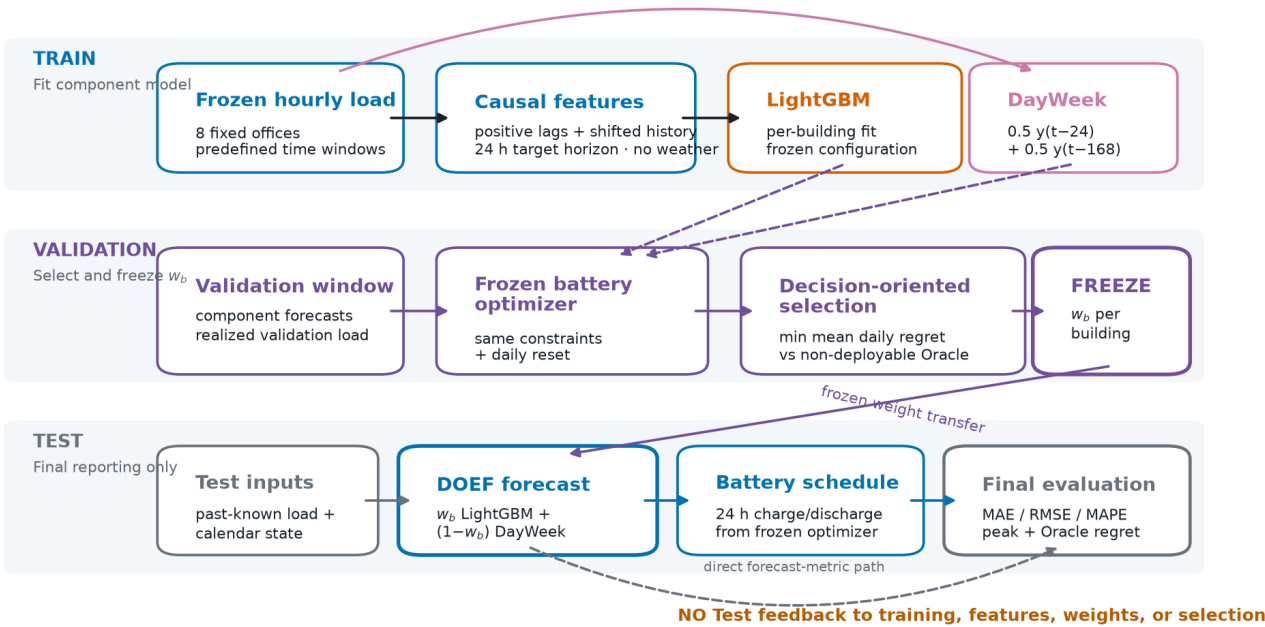
Appendix · open for “why only 8 / cherry-picking?”

因果特征与时间切分共同约束信息边界

24 h horizon ; 只使用预测时刻可获得的负荷历史与日历信息; Validation 选择, Test 最终评估。

DOEF v1.0 — frozen causal architecture

Prediction serves the storage decision · Validation selects · Test evaluates only



ALGORITHM FROZEN · deterministic ensemble and constrained optimization · not joint training, bilevel optimization, online learning, or a deployable Oracle.

信息边界

- positive lags
- shifted history
- calendar state
- no weather
- no target leakage

TEST ≠ pristine blind set

Frozen source: ARCH-01 · prior Test visibility caveat retained

基础模型是对照与组件，不是被包装的创新点

DOEF 只组合 DayWeek 与逐建筑 LightGBM ； 其他模型用于基线或受控对照。

Method	Definition / input	Role in benchmark	Novelty claim?
Day Persistence	t-24 h	Baseline	No
Week	t-168 h	Baseline	No
DayWeek	0.5·Day + 0.5·Week	DOEF component	No
LightGBM	causal features	DOEF component / baseline	No
XGBoost	causal features	Comparator	No
CatBoost	causal features	Comparator	No
DOEF	$w_{\beta} \cdot \text{LGBM} + (1 - w_{\beta}) \cdot \text{DayWeek}$	Final frozen method	Framework
Peak-aware LGBM	supporting experiment	Rejected support	No

Frozen benchmark definitions · no model-family catalogue in main deck

同一冻结电池约束保证方法间可比

所有预测方法进入完全相同的储能优化器；参数按每栋建筑 Train 期负荷尺度冻结。

Capacity 10% of mean daily energy kWh	Pcharge / Pdischarge 25% of capacity per hour kW
SOC range 10% → 90% kWh state	Initial / terminal SOC 50% / 50% daily reset
ηcharge / ηdischarge 0.94868 / 0.94868 round-trip 0.90	Optimizer deterministic daily peak shaving scipy.optimize.milp

1-hour interval: load energy kWh • dispatch power kW • capacity/SOC kWh

Frozen source: outputs/phase7/building_battery_configs.csv · same battery/optimizer for all forecast methods

完整预测指标用于核查，不替代下游决策评价

冻结八栋 Test 等权聚合； normalized metrics 均按每栋 Train mean load 归一化。

Method	normalized MAE	normalized RMSE	normalized regret	peak reduction mean
Day Persistence	.225885	.430193	.162986	-1.878%
Week	.201578	.328414	.144323	3.085%
DayWeek	.183478	.296621	.143019	4.907%
LightGBM	.147338	.216942	.116127	1.008%
XGBoost	.149781	.216751	.117534	0.453%
CatBoost	.173604	.239211	.118740	-24.482%
DOEF	.137970	.207822	.111929	6.442%
Peak-aware LGBM	.147587	.217284	.116202	1.129%

Canonical source: outputs/phase9/final_benchmark.csv · lower is better for normalized MAE/RMSE/regret

决策指标必须连同范围与聚合口径一起解释

mean 与 median 可能给出完全不同的直觉；负例必须和有利统计一起展示。

为什么不能只看一个平均数？

CatBoost peak reduction

-24.48% +4.82%

mean median

极端负例会强烈拖动 mean ； 因此报告必须同时给出 method / building / split / aggregation 。

min = -245.94%

Method	norm regret	peak mean	peak median	peak min
LightGBM	.116127	1.008%	5.057%	-41.192%
XGBoost	.117534	0.453%	4.588%	-44.357%
CatBoost	.118740	-24.482%	4.821%	-245.943%
DOEF	.111929	6.442%	5.629%	-1.017%

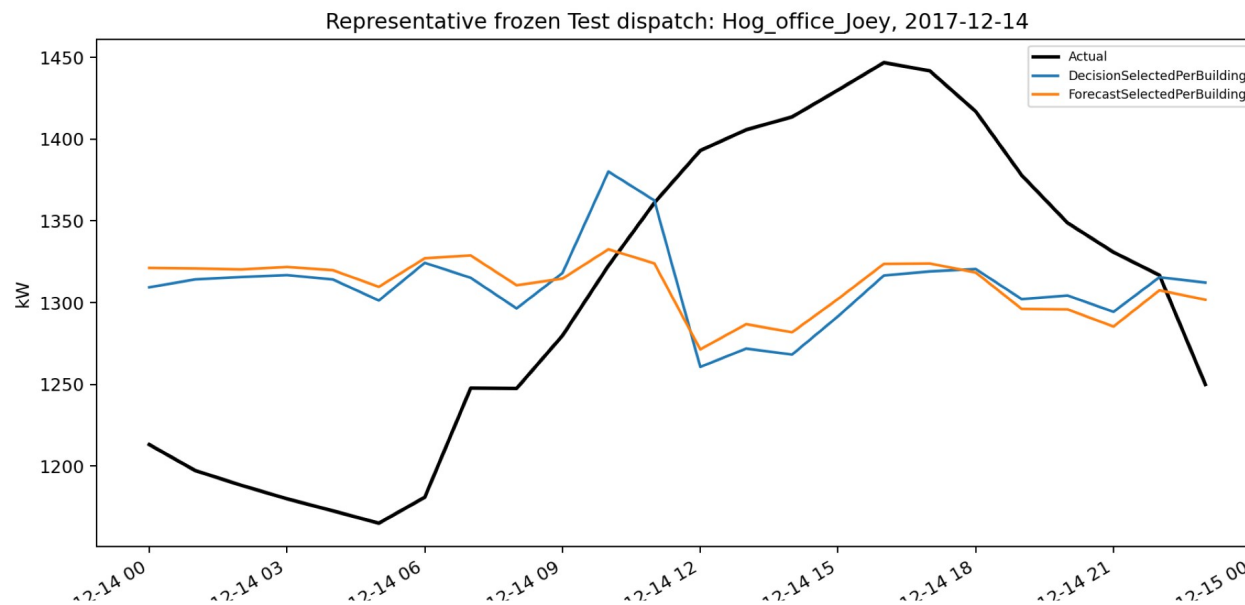
Frozen source: outputs/phase9/final_benchmark.csv + decision metrics · peak reduction is not a universal scalar

冻结调度轨迹只用于说明流程，不代表普遍表现

Phase 8 预先冻结的代表性 Test case；只作为“预测如何改变 realized grid load” 的机制说明。

Frozen representative forecast-to-dispatch trace

Phase 8 pre-frozen Test case · unchanged raster evidence · presentation frame only



Read as a mechanism trace, not a universal performance claim; the case and values remain frozen.

BACKUP ONLY

- 预冻结 Test case
- 不代表普遍表现
- 不做 case reselection
- 不提取新数值
- 不改变坐标 / 标签

Exact frozen source: outputs/phase13_1/figures/sci_05_storage_dispatch_frozen.png · no value extraction or case reselection

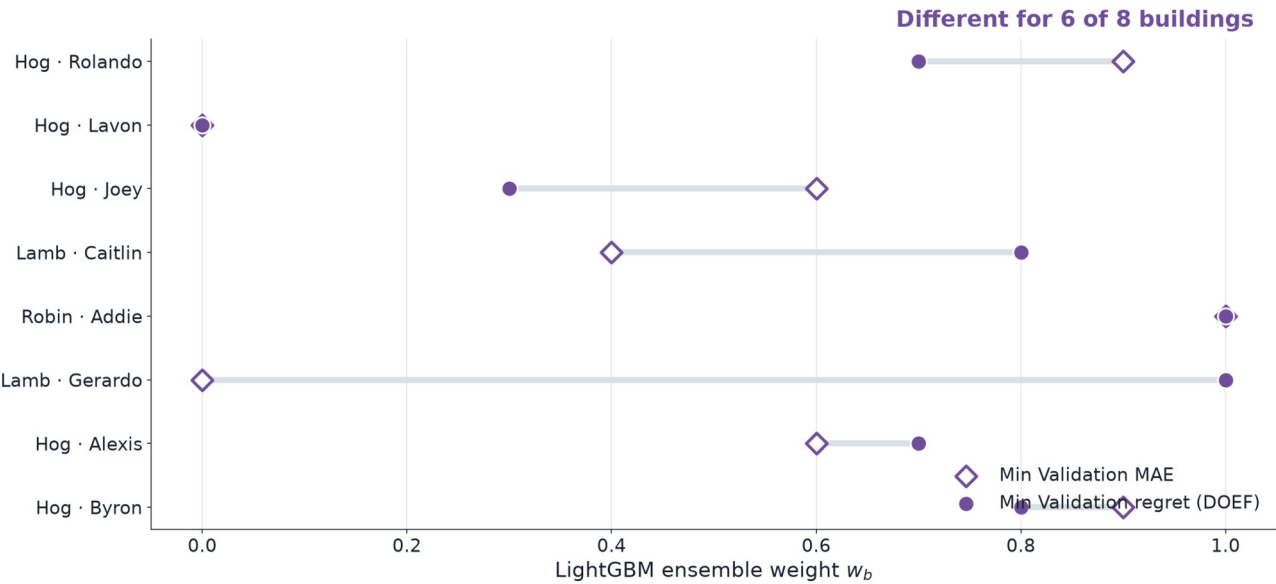
逐建筑权重来自预先声明的 Validation 搜索

候选 $w_\beta = \{0.0, 0.1, \dots, 1.0\}$; 确定性 tie-break ; Test 完全排除。

Joey: 0.6 → 0.3

Validation objectives select different ensemble weights

Per-building LightGBM weight; remaining weight is DayWeek · selection uses Validation only



Test is excluded from weight and tie-break selection. Points show frozen Phase 8 weights; no optimization was rerun.

Building	w forecast	w decision
Rolando	.9	.7
Lavon	0	0
Joey	.6	.3
Caitlin	.4	.8
Addie	1	1
Gerardo	0	1
Alexis	.6	.7
Byron	.9	.8

Frozen source: selected_weights.csv + SCI-04 · no interpolation between grid points

冻结 bootstrap 支持固定八栋建筑范围内的差异

paired building-level bootstrap ; resampling unit = building ; 不扩展为跨建筑普遍显著性结论。



95% interval **[-0.006850 , -0.001719]**

10,000

resamples · seed 42

-0.004198

point estimate

fixed 8-building scope only

报告公式与冻结预测逐值一致

这是 identity audit：显式 DOEF 公式复算与冻结 Phase 8 ensemble_prediction 逐值对齐，不是新训练或独立重复实验。

$$w_{\beta} \times \text{LightGBM} + (1 - w_{\beta}) \times \text{DayWeek}$$

=

frozen Phase 8
ensemble_prediction

IDENTITY AUDIT

5,760

frozen predictions checked

0.0 kW

max absolute difference

8 buildings × 720 Test hours

audited definition = frozen prediction artifact
within tolerance $1e-12$

负结果被保留，但不用于 Test 后反向改选算法

Validation-selected robust variant 在 Test 上明显退化，因此被保留为负结果并排除出 DOEF v1.0。

Validation selection

robust $\lambda = 0.5$

预先按 Validation 规则选出，随后进入 Test；没有因为 Test 结果差而换 λ 。

selection integrity

Test mean regret

20.4635 kW

robust variant

11.1274 kW

deterministic

结论：robust optimization 没有改善 Test，因此不进入 DOEF，不做 Test 后反向修正。

NEGATIVE RESULT RETAINED